

No Lipschitz Tube-Holomorphic Separator on a Banach Space Containing c_0

Automated mathematics run `fa_banach_001`, lane 17

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Status. Candidate full solution, likely valid, requiring human review. The theorem answers negatively the explicit c_0 version of the Lipschitz-extension hypothesis in source Theorem 2 and strengthens it to every space containing an isomorphic copy of c_0 .

Abstract

Azagra, Fry, and Keener ask whether c_0 has a Lipschitz real-analytic separating function of the kind used in their approximation theorem: its holomorphic extension is Lipschitz on a complex tube of fixed positive width. We prove that it does not. More generally, no real Banach space containing an isomorphic copy of c_0 admits such a separator. Continuous homogeneous polynomials have summable diagonal values on a c_0 -basic sequence. The Taylor expansion therefore makes the holomorphic extension vanish asymptotically along the corresponding coordinate lines near zero. Its global Lipschitz bound makes those line restrictions a normal family on one common strip, so the identity theorem propagates the convergence to every fixed real point. This contradicts linear separation at infinity.

1 Source question

The source is D. Azagra, R. Fry, and L. Keener, *Real analytic approximation of Lipschitz functions on Hilbert space and other Banach spaces*, arXiv:1005.1050v5, published in J. Funct. Anal. **262** (2012), 124–166 [1]. It calls $Q : X \rightarrow [0, \infty)$ a separating function if $Q(0) = 0$ and there are $m, M > 0$ such that

$$Q(x) \geq m\|x\| \quad (\|x\| \geq M).$$

Theorem 2 assumes that Q has a *Lipschitz holomorphic extension* $\tilde{Q}$ on a fixed-width tube

$$\Omega_\delta = \{x + iy \in X_{\mathbb{C}} : x, y \in X, \|y\| < \delta\}.$$

Immediately afterward, on PDF page 4, the authors write:

“We do not know if the converse is true. A natural related question is: does the space c_0 have such a Lipschitz separating function?”

We answer the explicit Theorem 2 version: the extension itself is globally Lipschitz on Ω_δ .

2 Proof intuition

Let (u_n) be a sequence equivalent to the canonical basis of c_0 . The key algebraic fact is that for every continuous homogeneous polynomial P , the diagonal sequence $(P(u_n))$ is summable. One

It is clear that X has a real analytic Lipschitz separating function (with a holomorphic extension to a uniformly wide neighborhood of X in $\tilde{X}$) if and only if X has a real analytic Lipschitz function Q which satisfies the above definition with $M = m = 1$ (and with a holomorphic extension to a uniformly wide neighborhood of X in $\tilde{X}$). So we also have the following.

Theorem 2. *Let X be a separable Banach space having a separating function $Q : X \rightarrow [0, \infty)$ with a Lipschitz holomorphic extension $\tilde{Q}$ defined on a set of the form $\{x + iy \in \tilde{X} : x, y \in X, \|y\| < \delta\}$ for some $\delta > 0$. Then there exists a number $C \geq 1$ such that for every Lipschitz function $f : X \rightarrow \mathbb{R}$, and for every $\varepsilon > 0$ there exists a Lipschitz, real analytic function $g : X \rightarrow \mathbb{R}$ such that $|f(x) - g(x)| < \varepsilon$ for all $x \in X$, and $\text{Lip}(g) \leq C \text{Lip}(f)$.*

We will prove (see Lemma 2 below) that every Banach space with a separating polynomial (that is a polynomial $P : X \rightarrow \mathbb{R}$ such that $P(0) = 0 < \inf\{P(x) : \|x\| = 1\}$) also has a Lipschitz, real analytic separating function with a holomorphic extension to a uniformly wide neighborhood of X in $\tilde{X}$. We do not know if the converse is true. A natural related question is: does the space c_0 have such a Lipschitz separating function?

Figure 1: Theorem 2 and the c_0 question in arXiv:1005.1050v5, PDF page 4.

sees this by inserting independent roots of unity into finite sums $\sum z_n u_n$: averaging kills all mixed monomials and retains only $P(u_n)$.

Now let F be the hypothetical holomorphic extension. Its Taylor expansion at zero and diagonal summability imply $F(tu_n) \rightarrow 0$ for small real t . The scalar functions $f_n(z) = F(zu_n)$ are holomorphic on one common horizontal strip. Because F is Lipschitz on the whole tube, they are uniformly bounded on compact subsets of that strip. Montel's theorem gives locally uniform subsequential limits; every limit vanishes on the small real interval and is therefore identically zero. Thus $F(Tu_n) \rightarrow 0$ for every fixed real T . Choosing T large contradicts the separator inequality, uniformly in n .

3 The diagonal polynomial lemma

We use the following form, which makes the later strengthening automatic.

Lemma 1 (summable diagonals). *Let E be a complex Banach space and let $(u_n) \subset E$ satisfy*

$$\left\| \sum_{n=1}^N z_n u_n \right\| \leq B \max_{1 \leq n \leq N} |z_n| \quad (1)$$

for all N and all complex scalars $z_1, \dots, z_N$. If $P : E \rightarrow \mathbb{C}$ is a continuous k -homogeneous polynomial, then

$$\sum_{n=1}^{\infty} |P(u_n)| < \infty.$$

For $k \geq 2$, in fact $\sum_n |P(u_n)| \leq B^k \|P\|$, where $\|P\| = \sup_{\|x\| \leq 1} |P(x)|$.

Proof. For $k = 1$, P is linear. Given N , choose unit scalars λ_n so that $\lambda_n P(u_n) = |P(u_n)|$. Equation (1) gives

$$\sum_{n=1}^N |P(u_n)| = P\left(\sum_{n=1}^N \lambda_n u_n\right) \leq B\|P\|.$$

Let $k \geq 2$ and let A be the symmetric k -linear form associated with P . Choose unit scalars λ_n such that $\lambda_n^k P(u_n) = |P(u_n)|$. Let $\xi_1, \dots, \xi_N$ be independent, uniformly distributed k th roots of unity. Expanding $P(\sum_n \lambda_n \xi_n u_n)$ in terms of A , a monomial survives expectation only if the multiplicity of every ξ_n is divisible by k . Since the total degree is k , the surviving monomials are precisely those whose k indices are all equal. Hence

$$\sum_{n=1}^N |P(u_n)| = \mathbb{E} P\left(\sum_{n=1}^N \lambda_n \xi_n u_n\right).$$

Taking absolute values and using (1) yields the claimed bound, uniformly in N . $\square$

4 Nonexistence theorem

Theorem 1. *Let X be a real Banach space containing an isomorphic copy of c_0 , and let $X_{\mathbb{C}}$ be a complexification. There do not exist $\delta, L, m, M > 0$ and a holomorphic function*

$$F : \Omega_{\delta} = \{x + iy \in X_{\mathbb{C}} : \|y\| < \delta\} \longrightarrow \mathbb{C}$$

such that:

- (i) F is L -Lipschitz on Ω_{δ} ;
- (ii) $Q = F|_X$ is real-valued, $Q(0) = 0$, and $Q \geq 0$;
- (iii) $Q(x) \geq m\|x\|$ whenever $\|x\| \geq M$.

Proof. Choose an isomorphic embedding $T : c_0 \rightarrow X$ and put $u_n = Te_n$. After complexification there are constants $a, b > 0$ such that

$$\|u_n\| \geq a, \quad \left\| \sum_{n=1}^N z_n u_n \right\| \leq b \max_n |z_n| \quad (2)$$

for real vectors in the first inequality and complex finite scalar families in the second.

Write the Taylor expansion of F at zero as

$$F(w) = \sum_{k=0}^{\infty} P_k(w),$$

where P_k is a continuous complex k -homogeneous polynomial and $P_0 = F(0) = 0$. Since F is holomorphic and continuous at zero, there is a ball about zero on which it is bounded, and its Taylor series converges uniformly on every sufficiently smaller ball. Choose $\rho > 0$ so small that the series converges uniformly whenever $\|w\| \leq 2b\rho$.

Fix a real t with $|t| \leq \rho$. Lemma 1, applied to (2), gives $P_k(u_n) \rightarrow 0$ for every fixed k . Thus every fixed finite Taylor sum at tu_n tends to zero. The uniform Taylor tail on $\|tu_n\| \leq b\rho$ is independent of n , so

$$F(tu_n) \longrightarrow 0 \quad (|t| \leq \rho). \quad (3)$$

For every n , define $f_n(z) = F(zu_n)$. Since $\|u_n\| \leq b$, all f_n are holomorphic on the common strip

$$S = \{z \in \mathbb{C} : |\operatorname{Im} z| < \delta/b\}.$$

The Lipschitz hypothesis and $F(0) = 0$ give, for $z \in S$,

$$|f_n(z)| \leq L\|zu_n\| \leq Lb|z|. \quad (4)$$

Therefore (f_n) is a normal family on S . Take any subsequence. Montel’s theorem provides a further subsequence converging locally uniformly to a holomorphic function f on S . By (3), f vanishes on $[-\rho, \rho]$, and the identity theorem gives $f \equiv 0$ on S .

It follows that the full sequence satisfies $f_n(z) \rightarrow 0$ at every fixed $z \in S$: otherwise a subsequence bounded away from zero at some z would have a locally uniformly convergent subsubsequence, whose limit we just proved must be zero.

Finally choose a fixed real $R \geq M/a$. Then $\|Ru_n\| \geq aR \geq M$, and the separating inequality gives

$$f_n(R) = Q(Ru_n) \geq m\|Ru_n\| \geq maR > 0$$

for every n . This contradicts $f_n(R) \rightarrow 0$. □

Corollary 1. *The space c_0 has no separating function whose holomorphic extension is globally Lipschitz on a complex tube of fixed positive width. Thus the answer to the explicit Theorem 2 version of the source question is negative.*

5 Scope, comparison, and novelty status

Why the classical c_0 separator does not conflict. The function

$$d(x) = \sum_{n=1}^{\infty} x_n^{2n}$$

is a standard uniformly analytic separating function on c_0 [2, 4]. It is not globally Lipschitz: on the coordinate line te_n , its derivative has size $2n|t|^{2n-1}$. The theorem concerns the strictly stronger Lipschitz-tube hypothesis needed in source Theorem 2.

Interpretive limitation. The source’s Theorem 2 explicitly says that the *holomorphic extension* is Lipschitz on the tube, and this is the property proved impossible here. If the phrase “such a Lipschitz separating function” were instead intended to require only that Q be Lipschitz on the real space while allowing its tube extension to have arbitrary growth, estimate (4) would no longer follow and that weaker variant remains outside this packet.

Bounded novelty audit. The run indexes contained no duplicate. OpenAlex listed thirteen citing works for the published source as of 11 August 2026. Their titles and the available relevant full texts were checked. Johani’s simpler proof [3] retains the separating-polynomial hypothesis. Mytrofanov’s survey [4] and a 2021 symmetric-approximation paper discuss the weaker uniformly analytic c_0 separator above, not a globally Lipschitz tube extension. Exact-phrase and close-variant searches found no direct answer. Novelty confidence is moderate-to-high under this bounded audit, not an exhaustive priority claim.

Human-review recommendation. Prioritize review. Check the root-of-unity extraction in Lemma 1, the normal-family step, and the intended interpretation of the source’s word “such.” No conditional mathematical dependency remains for the explicit Theorem 2 version.

References

- [1] D. Azagra, R. Fry, and L. Keener, *Real analytic approximation of Lipschitz functions on Hilbert space and other Banach spaces*, J. Funct. Anal. **262** (2012), 124–166; arXiv:1005.1050v5.
- [2] M. Cepedello Boiso and P. Hájek, *Analytic approximations of uniformly continuous functions in real Banach spaces*, J. Math. Anal. Appl. **256** (2001), 80–98.
- [3] M. Johanis, *A simple proof of the approximation by real analytic Lipschitz functions*, J. Math. Anal. Appl. **388** (2012), 1–7.
- [4] M. A. Mytrofanov, *Separating polynomials, uniform analytical and separating functions*, Carpathian Math. Publ. **7** (2015), 197–208.